



UNIVERSITY OF TARTU



Enterprise System Integration (MTAT.03.229)

COURSE INTRODUCTION

UNIVERSITY OF TARTU

Slides created by Mohamad Gharib,
and adapted by Orlenys López Pintado
with the content of the current semester

Course description

The **objectives** of this course is to introduce principles and methods for designing, developing, and integrating enterprise information systems, including:

- **Application programming interface (API)** design methods and patterns of enterprise integration;
- **Representational state transfer (REST)** architectural, RESTful API - (CRUD operations), etc.;
- **Service-oriented** and **Microservices architectures** and their associated technologies.
- Frameworks & tools for implementing & integrating enterprise applications.



Who are we?

Orlenys López Pintado

Lecturer

Iwada Eja Bassey

Supervisor of practical sessions



Course structure

- **10 Lectures** (W1-W10),
 - **1 lecture** for project introduction and exam method description (W11),
 - **2 lectures** for project consultation – on demand (W12-13),
 - **Assignments** 1, 2, and 3 will be discussed with me.
-
- **10 Practical sessions** (W1-W10).
 - **Project:** 3 checkpoints + 2 session for consultation (W11-W15).

See details on the Wiki page: <https://courses.cs.ut.ee/2026/esi>

Course structure

Lectures

**Practical
sessions**

Lectures

Why, What, How

Practical sessions

Why, What, **How**

Course structure

Lectures	Essential technologies for ESI	DDD -> microservices arch	Vue.js	Security Aspects of Integration
Practical sessions	Spring Boot for SOA	Microservices -Spring Boot		

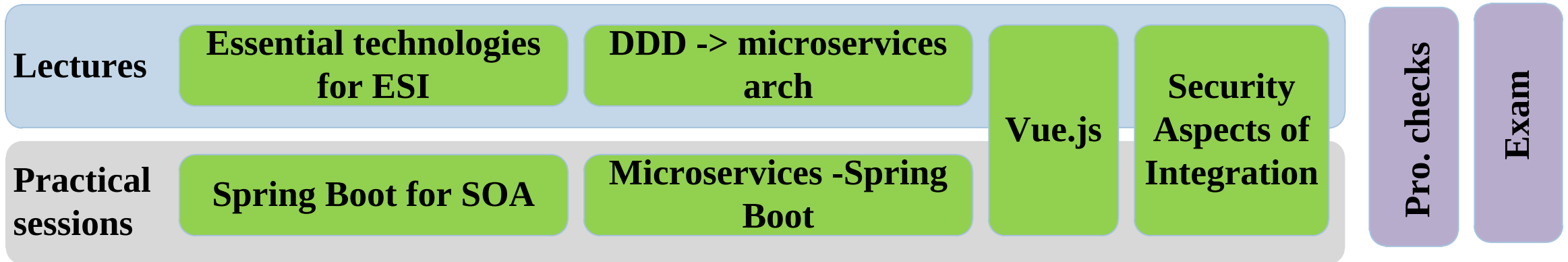
Lectures

Why, What, How

Practical sessions

Why, What, **How**

Course structure



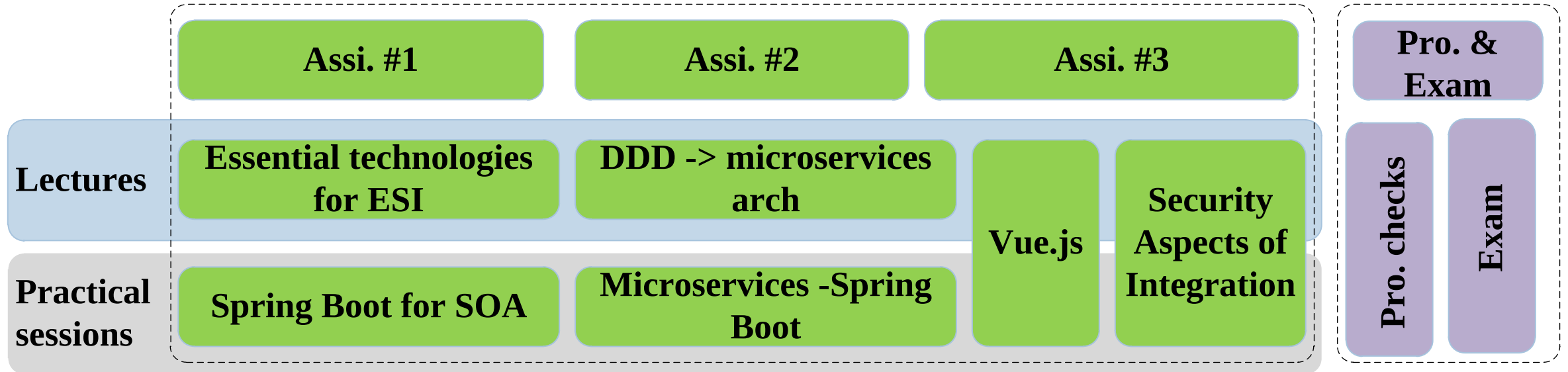
Lectures

Why, What, How

Practical sessions

Why, What, **How**

Course structure



Lectures

Practical sessions

Why, What, How

Why, What, **How**

Assignments #1, 2 & 3 – rely on Problem-Based Learning (PBL)

https://en.wikipedia.org/wiki/Problem-based_learning

Course content in details

Course introduction

Introduction to the ESI
course

Enterprise System
integration and
interoperability

Web Services

Representational state
transfer (REST)
architecture

RESTful API (CRUD
operations)

System Engineering

Requirements
Engineering

[Enterprise] System
Architecture

System Modeling

Conceptual/Meta/
Domain Modeling

Lectures

Background

Lab

Course content in details

Course introduction

Introduction to the ESI course

Enterprise System integration and interoperability

Web Services

Representational state transfer (REST) architecture

RESTful API (CRUD operations)

System Engineering

Requirements Engineering

[Enterprise] System Architecture

System Modeling

Conceptual/Meta/
Domain Modeling

From monolithic to microservices architecture

Theory

The Microservices Architecture

The Domain-driven design (DDD) Approach

Domain-Oriented Microservice Architecture

Integration patterns in Microservice architecture

Practice

Spring Boot - Microservices – I

Spring Boot - Microservices – II

Spring Boot - Microservices – III

Spring Boot - Microservices integration patterns

Lectures

Background

Lab

Course content in details

Course introduction

Introduction to the ESI course

Enterprise System integration and interoperability

Web Services

Representational state transfer (REST) architecture

RESTful API (CRUD operations)

System Engineering

Requirements Engineering

System Modeling

[Enterprise] System Architecture

Conceptual/Meta/Domain Modeling

From monolithic to microservices architecture

Theory

The Microservices Architecture

The Domain-driven design (DDD) Approach

Domain-Oriented Microservice Architecture

Integration patterns in Microservice architecture

Practice

Spring Boot - Microservices – I

Spring Boot - Microservices – II

Spring Boot - Microservices – III

Spring Boot - Microservices integration patterns

Designing and implementing frontend apps

Vue.js

Lectures

Background

Lab

Course content in details

Course introduction

Introduction to the ESI course

Enterprise System integration and interoperability

Web Services

Representational state transfer (REST) architecture

RESTful API (CRUD operations)

System Engineering

Requirements Engineering

System Modeling

[Enterprise] System Architecture

Conceptual/Meta/Domain Modeling

From monolithic to microservices architecture

Theory

The Microservices Architecture

The Domain-driven design (DDD) Approach

Domain-Oriented Microservice Architecture

Integration patterns in Microservice architecture

Practice

Spring Boot - Microservices – I

Spring Boot - Microservices – II

Spring Boot - Microservices – III

Spring Boot - Microservices integration patterns

Designing and implementing frontend apps

Vue.js

The Security Aspects of Integration

Theory

Security Aspects of Integration - I - basic concepts

Security Aspects of Integration - II – advanced concepts

Practice

Security Aspects of Integration - I - basic concepts

Security Aspects of Integration - II – advanced concepts

Lectures

Background

Lab

Course content in details

Course introduction

Introduction to the ESI course

Enterprise System integration and interoperability

Web Services

Representational state transfer (REST) architecture

RESTful API (CRUD operations)

System Engineering

Requirements Engineering

System Modeling

[Enterprise] System Architecture

Conceptual/Meta/Domain Modeling

From monolithic to microservices architecture

Theory

The Microservices Architecture

The Domain-driven design (DDD) Approach

Domain-Oriented Microservice Architecture

Integration patterns in Microservice architecture

Practice

Spring Boot - Microservices – I

Spring Boot - Microservices – II

Spring Boot - Microservices – III

Spring Boot - Microservices integration patterns

Designing and implementing frontend apps

Vue.js

The Security Aspects of Integration

Theory

Security Aspects of Integration - I - basic concepts

Security Aspects of Integration - II – advanced concepts

Practice

Security Aspects of Integration - I - basic concepts

Security Aspects of Integration - II – advanced concepts

Project consultation/checkpoints, and Exam preparation

Project consultation

Exam method description

Project checkpoints

Project presentations

Lectures

Background

Lab



Grading

Assignments: 20 points

Project: 30 points.

Exam: 50 points (minimum of **21/50 required to pass the exam).**

The resulting sum (out of 100) will be mapped to a grade from **A to F** (minimum of **51/100** required to pass the exam).

Two exams will be offered, and **one resit exam** will be offered to students who **fail** or **cannot attend any of the two exams**.

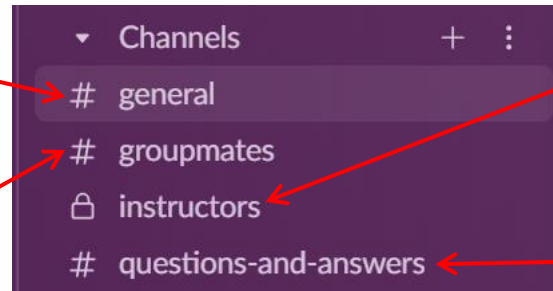
Course rules and communication

- You will be split into teams of 4 members, you need to register your team by the end of next week ([Via Moodle](#)).
- Assignments and project deliverables are teamwork, yet each member of the team is expected to answer any question about the work, and they will be discussed by All team members.
- Points will be granted individually not as a team. Optimally, all the team members will get the same points.
- Do not forget to join the Slack Workspace ([Link](#))  **slack**

Course rules and communication

Why you should join Slack?

General channel: contains announcement, important information, etc.



You do not have access to this one 😊

Q&A: list a common question and receive an answer.

Groupmates: helps to connect students trying to find their group mates.

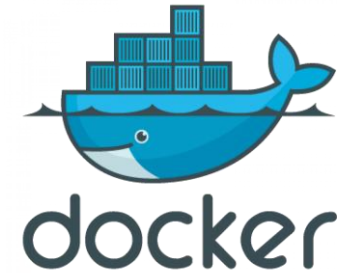


Join Here !!!

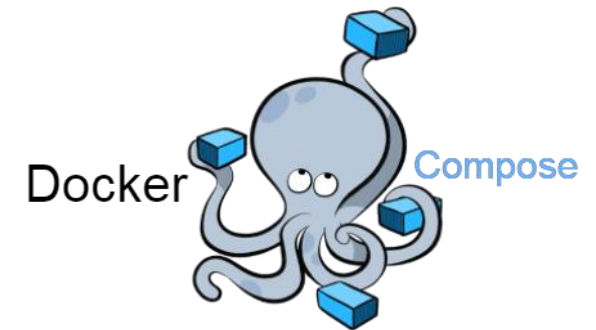
IDEs, framework, tools, etc.



VS Code



IntelliJ





How to study

- Attend the **lectures**.
- Attend the **practical sessions**, and do the related **tasks**.
- Do the **assignments on time**.
- Do the **practical examples**.
- Read the additional **resources**.
- **Apply everything you learn**.

Final Note

Experience has shown that your final grade is highly correlated with lectures attendance.



UNIVERSITY OF TARTU

**Thank You
for your attention**

Orlenys López Pintado
orlenys.lopez.pintado@ut.ee



unitartu



tartuuniversity

